Major Project-I · CSEE-712P

IndexR

A Vectorless Research Assistant with Section- and Page-Level Cited Answers

Bachelor of Technology in Computer Science & Engineering (AI&ML) · 7th Semester

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Problem Statement

Find the right papers for a research question, answer only from them with every claim cited to the exact paper, section and page, and keep the topic up to date, using lexical retrieval instead of a vector database.

1

Fragmented discovery

Papers are spread across arXiv, OpenAlex, Semantic Scholar; most tools expect manual uploads

2

Unverifiable answers

LLMs cite nothing, cite whole documents, or invent references

3

Costly vector pipelines

Embeddings, a separate vector database and re-indexing on model change

4

Loss of exact terms

Embeddings blur model names, dataset names and numbers like “F1 = 0.95”

5

Stale knowledge

Nothing reports new papers, new citations or new versions on a topic

6

Isolated from AI tools

The research corpus is not available inside AI assistants and editors

Objectives

O1

Shared, de-duplicated corpus from arXiv, OpenAlex and Semantic Scholar with section/page chunks

O2

Vectorless retrieval: full-text search, citation and recency boost, query expansion, reranking

O3

Grounded, streamed answers whose [n] markers map to paper, section and page

O4

Two-gate rule that decides when to discover new papers

O5

Responsive web app for workspaces, chat, papers and settings

O6

Private PDF uploads visible only to their owner

O7

MCP server with 11 tools and token authentication for AI assistants

O8

Live tracking of new papers, new citations and new arXiv versions

O9

Thorough validation: 400+ automated tests and constant-query list endpoints

Literature Review and Research Gap

Area	Key work	What it shows
Lexical retrieval	BM25 [1]; BEIR [5]	BM25 is a robust zero-shot baseline; BM25 + reranker is strong
Dense retrieval	DPR [3]; ColBERT [4]; HNSW [26]	Effective, but needs embeddings and a vector index
Reranking	monoBERT [6]; Lin et al. [7]	Cross-encoders are accurate over a short candidate list
RAG	Lewis et al. [2]; Self-RAG [10]	Grounding helps; deciding when to retrieve matters
Citations	ALCE [9]; attributed QA [20]	LLM citations are often coarse or unsupported
Scholarly data	OpenAlex [16]; Semantic Scholar [17]	Open metadata, citation and version data

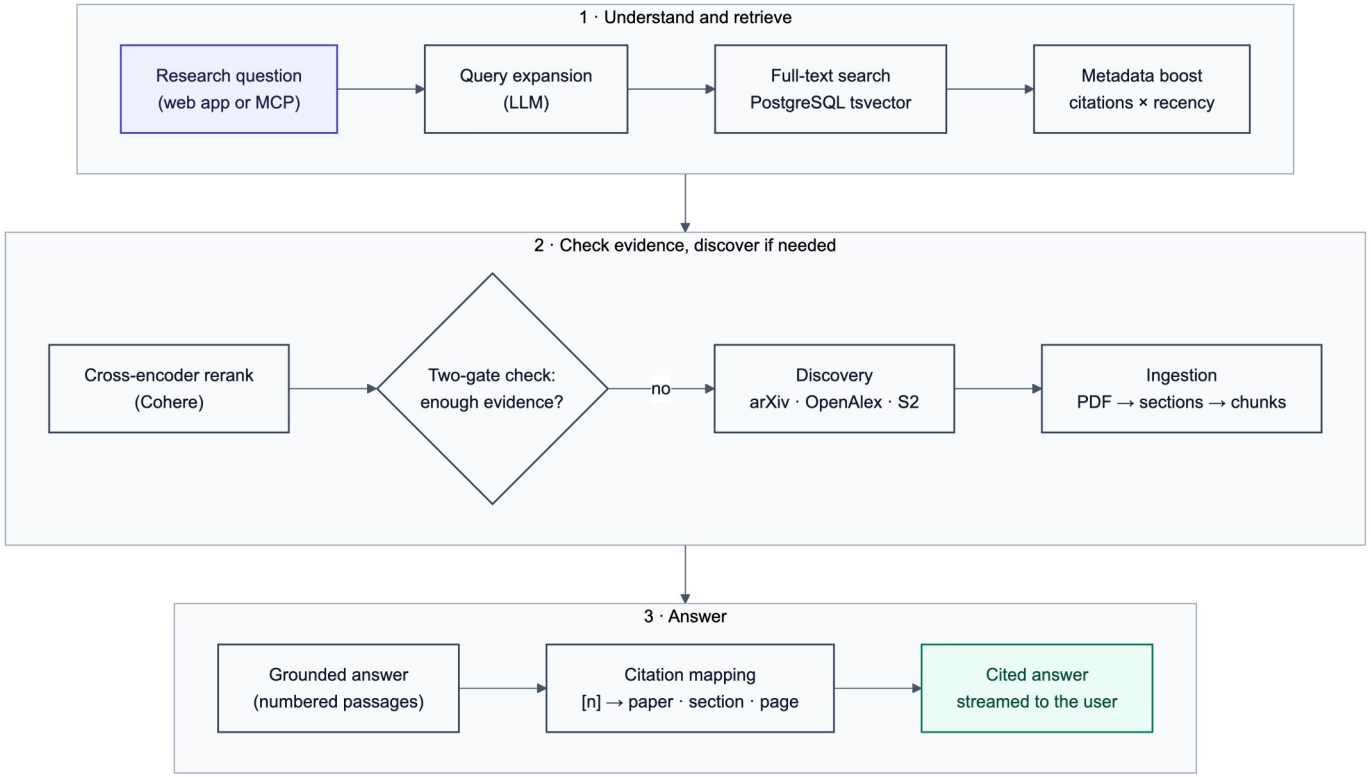
Research gap

- G1 RAG assumes a vector database
- G2 Citations are coarse or unreliable
- G3 Corpora are static and user-supplied
- G4 No awareness of new work
- G5 Corpus isolated from AI assistants

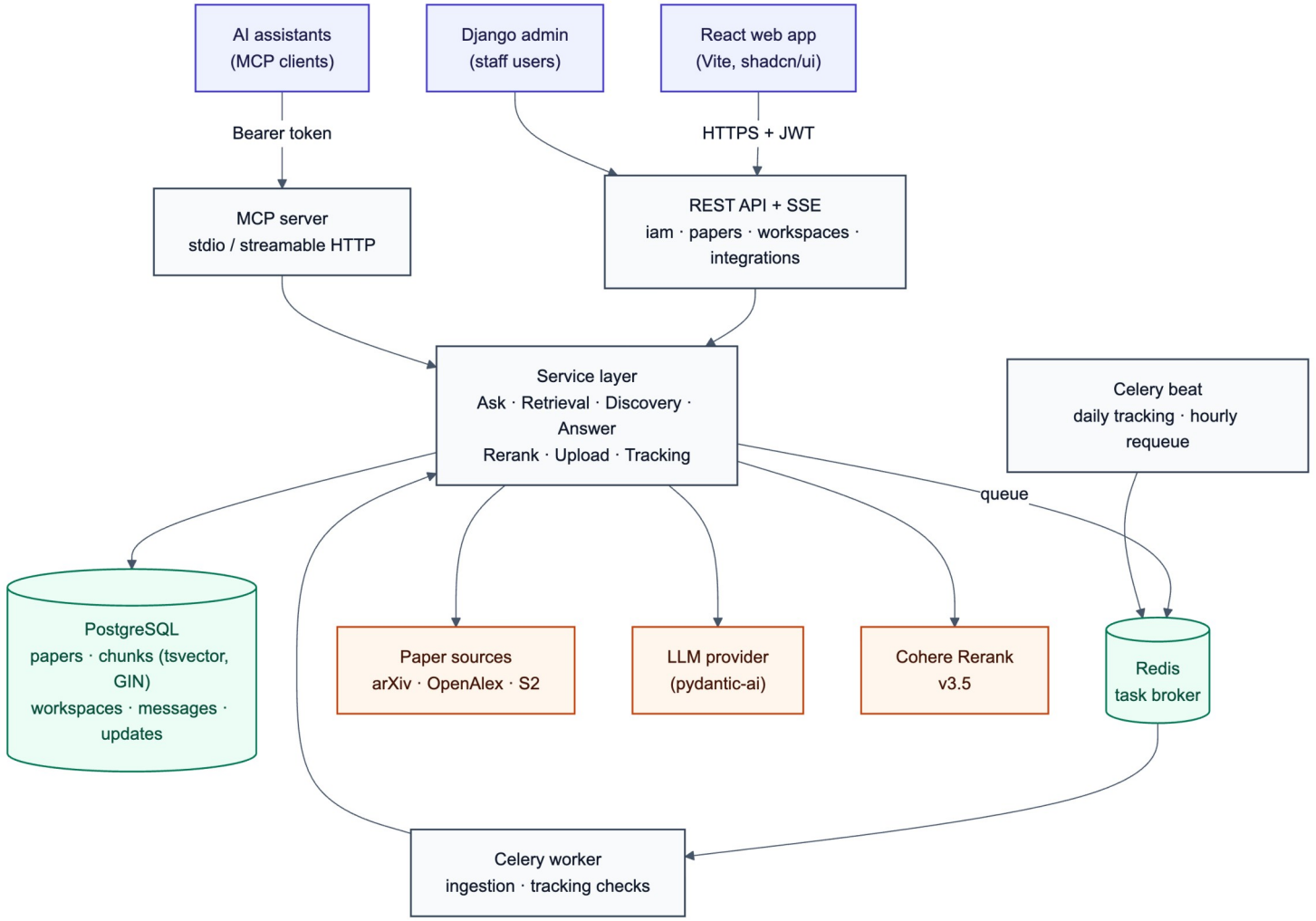
IndexR addresses all five in one system.

Proposed Approach: Vectorless Retrieval

- Section-aware chunks with page ranges, indexed with PostgreSQL full-text search (GIN)
- Scores boosted by citation count and recency
- LLM expands each question into 3–5 keyword queries
- Cohere cross-encoder reranks the top 20 candidates
- Two-gate rule triggers discovery when evidence is weak
- Answers cite numbered passages; locations come from the database

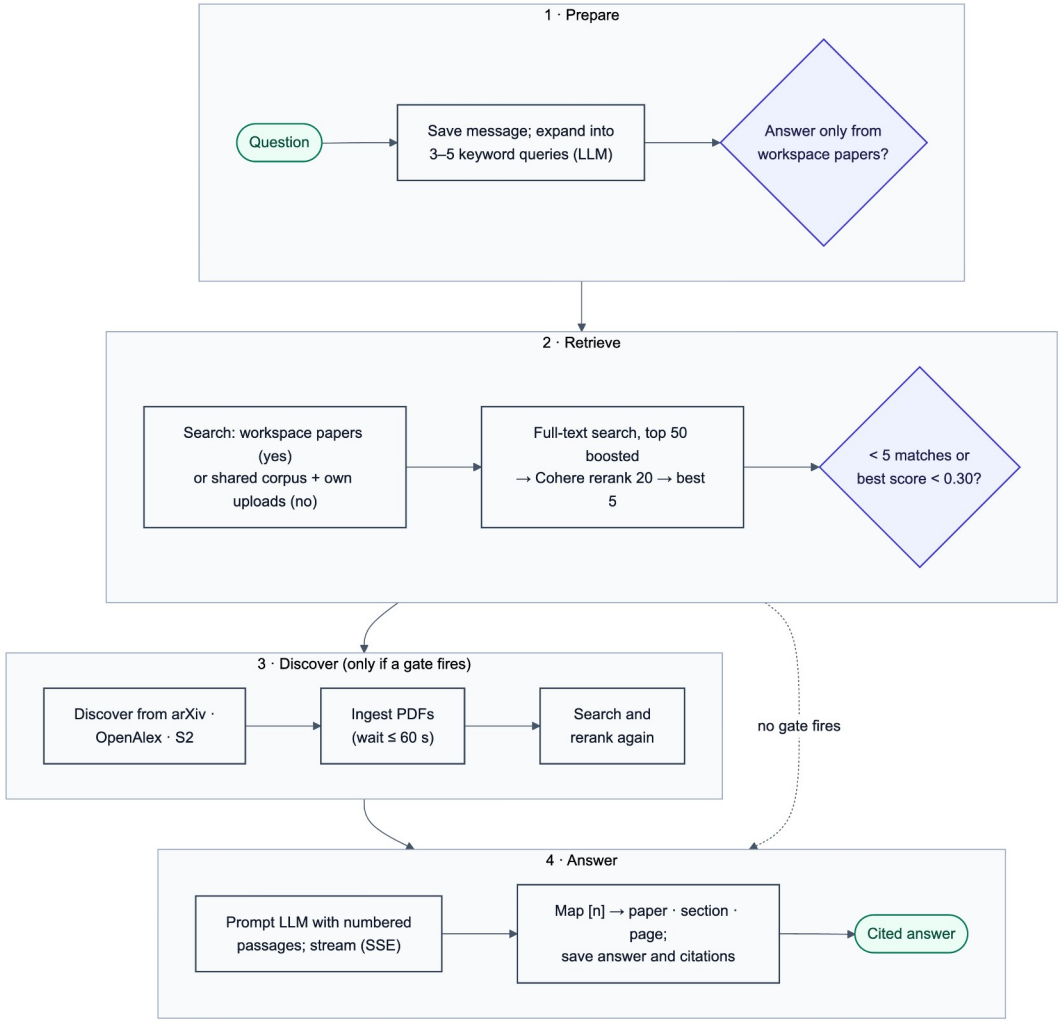


System Architecture



- React web app (JWT) and AI assistants (MCP, access tokens) share one service layer
- Django REST Framework API with streamed answers (SSE)
- Celery worker and beat for ingestion, tracking and requeueing
- PostgreSQL stores data and the full-text index; Redis is the broker
- External: arXiv, OpenAlex, Semantic Scholar, LLM provider, Cohere Rerank

Answering a Question



Two-gate discovery rule

Gate	Discover when
1. Count	Fewer than 5 chunks match
2. Confidence	Best rerank score below 0.30

Pipeline stages

- Understanding the question (query expansion)
- Looking through papers (search + rerank)
- Finding and reading new papers (only if a gate fires)
- Writing the answer, streamed live with status updates

Ranking, Ingestion and Citations

Ranking

$$\text{score} = \text{rank} \times \text{B_cit} \times \text{B_rec}$$

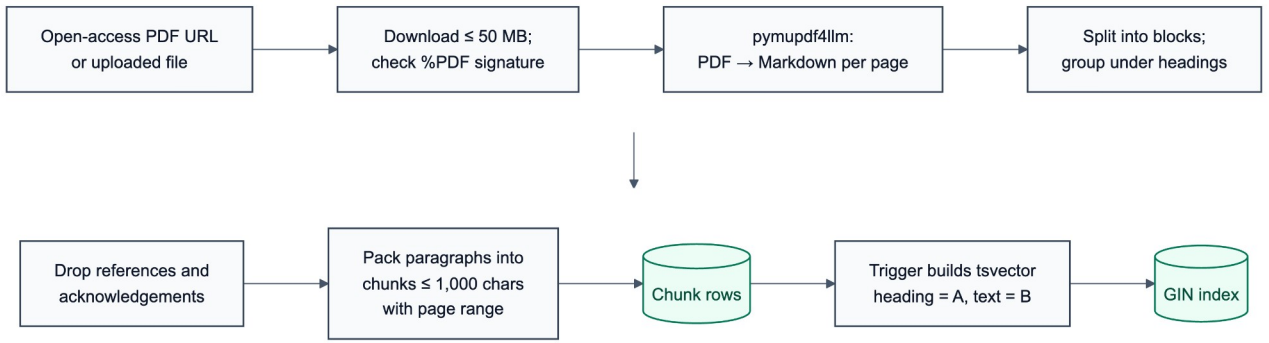
$$\text{B_cit} = 1 + 0.1 \times \log_{10}(\text{citations} + 1)$$

$$\text{B_rec} = 1 + 0.2 \times \max(0, 1 - \text{age} / 5)$$

Citations

- Top 5 passages numbered [1]–[5]; model may cite only these
- Each [n] mapped to its stored chunk; unknown numbers dropped
- Section and page shown come from the database, not the model

Ingestion and section-aware chunking



Live Tracking and MCP Integration

Live tracking (daily, Celery beat)

- New papers on tracked topics (published since the last check)
- New papers citing a tracked paper (OpenAlex)
- New arXiv versions of a tracked paper
- Each update reported once per workspace
- “N updates since date” banner and home-page badge

Limits

- Up to 10 topics per workspace; first check looks back 30 days
- At most 10 newest citing papers per paper per check
- Run on demand with manage.py check_tracking

MCP server (11 tools)

```
search_papers
ask
list_workspaces
create_workspace
list_workspace_papers
get_paper
add_paper_to_workspace
remove_paper_from_workspace
track_topic
untrack_topic
list_updates
```

Personal access tokens (SHA-256 hashed); every tool acts only as the token’s user.

Tools and Technologies

Backend

Python 3.12, Django 6.1, Django REST Framework, Simple JWT, django-allauth

Data

PostgreSQL 18 (tsvector, GIN index), Redis 8

Background jobs

Celery 5.6 worker and beat scheduler

AI services

pydantic-ai (Gemini / Claude / GPT), Cohere Rerank v3.5, pymupdf4llm

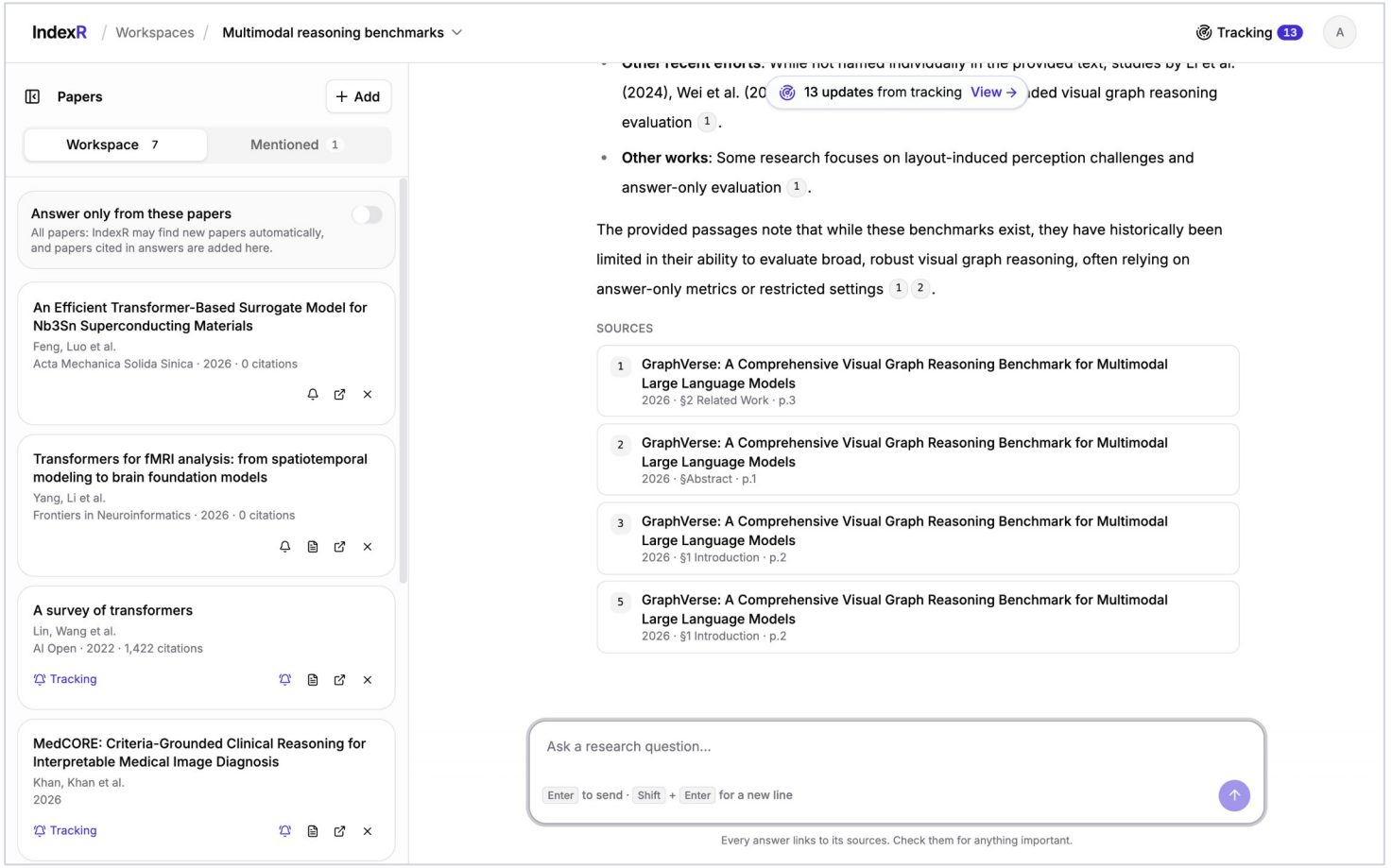
Frontend

React 19, TypeScript, Vite, TanStack Query, Tailwind CSS, shadcn/ui

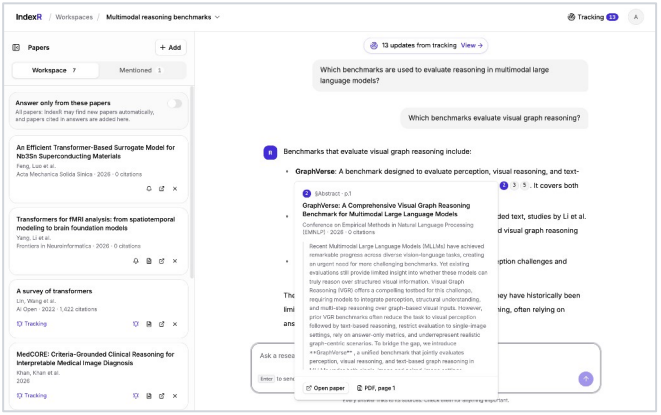
Integration and testing

MCP Python SDK, Django test runner, Playwright

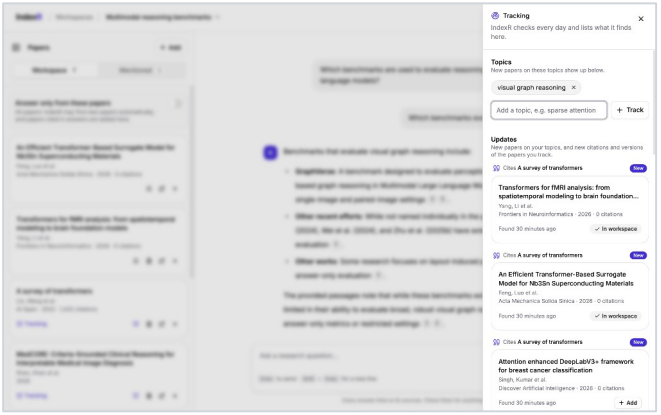
User Interface



Workspace: papers panel and a cited answer with its sources



Citation preview: section, page and passage



Tracking panel with topics and updates

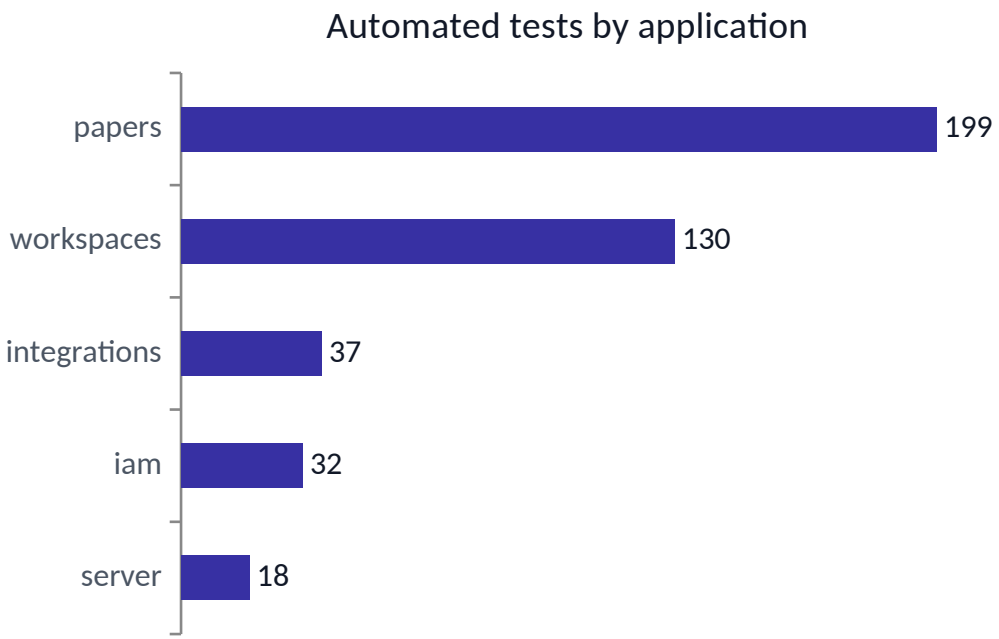
Validation and Results

416
automated tests, all passing

12 / 12
functional requirements verified

2.2 ms
median full-text search

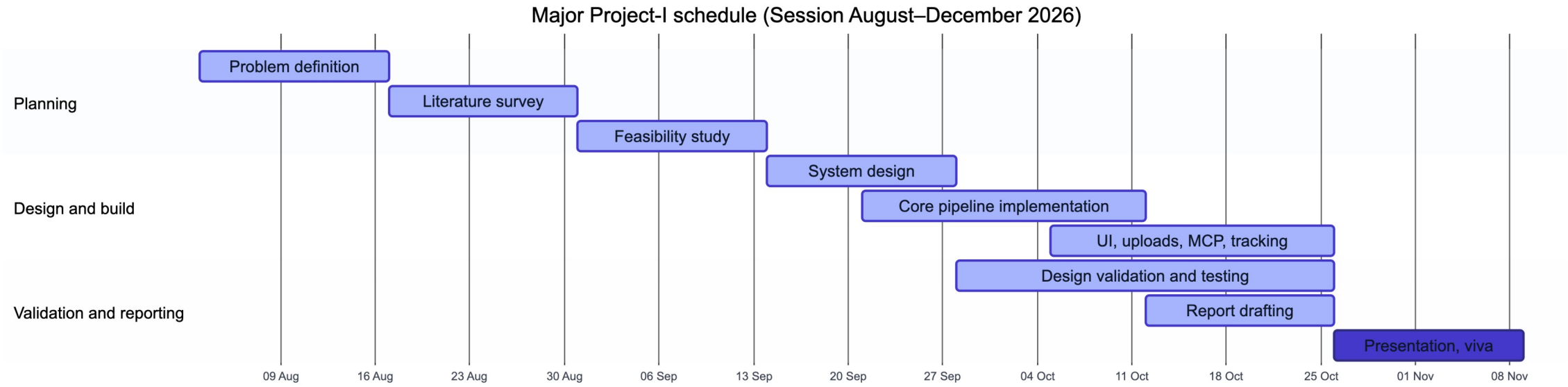
≈ 11 s
answer from the corpus



Live validation

- Corpus: 137 papers, 4,930 chunks
- MCP: requests without a valid token rejected (HTTP 401)
- Tracking: 3 new papers on a topic; 10 new citing papers; re-runs add no duplicates
- Every paginated list: same query count for 1 and 5 rows
- Two defects found by live runs, fixed with regression tests

Timeline



Weeks 1–6

Problem, literature survey, feasibility, setup and authentication

Weeks 7–10

Design, fetchers, ingestion, retrieval, answers, web app

Weeks 11–12

Uploads, MCP server, live tracking, validation, report

Weeks 13–14

Presentation and viva

Conclusion and Future Scope

Conclusion

- Cited, grounded answers without a vector database
- Automatic discovery from three open sources
- Citations traceable to section and page
- Private uploads, MCP integration and live tracking
- Backed by 416 tests and live validation

Future scope (Major Project-II)

- Formal evaluation on 20–30 questions
- Citation-graph expansion and visualisation
- Table, figure and equation extraction
- Version-aware re-ingestion
- Email digests and LLM summaries of updates
- Shared workspaces and organisation accounts

Lexical retrieval with query expansion and reranking is a practical, inexpensive and explainable foundation for trustworthy, cited answers.

Thank you. Questions?

References

[1] S. Robertson and H. Zaragoza, “The probabilistic relevance framework: BM25 and beyond,” Found. Trends Inf. Retr., 2009.

[2] P. Lewis et al., “Retrieval-augmented generation for knowledge-intensive NLP tasks,” NeurIPS, 2020.

[3] V. Karpukhin et al., “Dense passage retrieval for open-domain question answering,” EMNLP, 2020.

[5] N. Thakur et al., “BEIR: A heterogeneous benchmark for zero-shot evaluation of IR models,” NeurIPS Datasets & Benchmarks, 2021.

[6] R. Nogueira and K. Cho, “Passage re-ranking with BERT,” arXiv:1901.04085, 2019.

[9] T. Gao et al., “Enabling large language models to generate text with citations,” EMNLP, 2023.

[10] A. Asai et al., “Self-RAG: Learning to retrieve, generate, and critique through self-reflection,” ICLR, 2024.

[16] J. Priem, H. Piwowar, and R. Orr, “OpenAlex: A fully-open index of scholarly works,” arXiv:2205.01833, 2022.

[22] Anthropic, “Model Context Protocol specification,” modelcontextprotocol.io.

[26] Y. A. Malkov and D. A. Yashunin, “Efficient and robust approximate nearest neighbor search using HNSW graphs,” IEEE TPAMI, 2020.

Full list of 27 references in the project report.